SafGuard®

Product Mechanical Safety Testers











Safeguard Your business !

There are three key Product Safety Tests: Chemical, Flammability, and Mechanical Safety. Usually, the former two tests are needed to test once. However, Mechanical Safety is required to be tested in the process of product design to production mainly because it involves a lot of different workmanship in the process.
In order to meet more and more stringent standards & regulations as well as requirements by major retailers in mechanical safety, SafGuard can do for you to supply with a complete line of Mechanical Safety Testers mainly used for Children Garment, Footwear & Toys.

Total Product Mechanical Safety Solution

A proactive process, illustrated below, is the basic of your product integrity policy:

Design & Materials Selection

pass

1 Preliminary Test

Universal Mechanical Safety Tester
To test attachment strength of accessories

- **Torque Test Kit** — *To test torque strength of accessories*
- **Button Impact Tester** — *To test whether plastic buttons are easily breakable*
- **Needle or Metal Detector** — *To check whether broken needles or metal contaminated*
- **Zipper Test System (eg. Slide Fasteners, Strength, etc)** — *To test how durable a zipper is used*

fail

Product's accessories & packing material adopted are required to be tested before mass production.

pass

2 Assessment

- **Small Part Cylinder** — *To define a part or breakable part if it is a small part*
- **Sharp Point Tester** — *To define whether a part of accessory is a sharp point*
- **Sharp Edge Tester** — *To define whether an edge of accessory is a sharp edge*

fail

pass

3 Regular Checking in Production

0.1% random sampling for destructive tests

100% inspection for non-destructive tests / check

fail

pass

Shipment

Planning is as natural to the process of success as its absence is the process of failure.

Universal Mechanical Safety Tester

Attachment Strength Tests For All Accessories (e.g. Snaps, Zippers, & Buttons...)

This tester is the unique safety test instrument that provides a means to accurately measure how safe accessories are attached in a product. It is also used to compare a variety of accessories from different vendors in the process from product design to production so that we can safeguard our consumers.

It is a simple but effective design mainly consisting of a high precision gauge, a sturdy stand, and special grips/clamps. It requires very little training. A sample required by standards & regulations is to fix horizontally (constant force is applied to secure the sample) so that snap or other accessories attached in fabric can be clamped into.

Moreover, a standard weight is also offered so that it can be regularly checked if it meets the safety requirements.

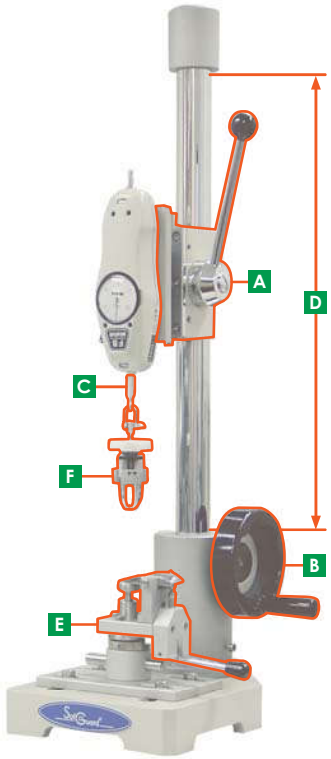


Standards Reference

BV, JAPAN Product Liability Law, ASTM F963, EN 71, M&S P115A (for snap), P115 (for button), ASTM D4846-96 And many others.

Test Interpretation

There are lots of different test interpretations. Take snaps as an example. All snaps are usually tested to withstand attachment strength of 17 pounds (or 7.73 kg) for 10 seconds. If not, it is deemed to be mechanically unsafe unless it is not a small part & there is no sharp edge/point...



SafGuard's Main Outstanding Features & Benefits:

Better Cost-Effective And Quality Design:

A **Ergonomically Constructed**
force meter holder can be retained upwards during sampling

B **User-friendly**
side-wheel operation

C **Economically exchangeable** with a plenty of different of grips/clamps.

D **Longer Stand Spacious** for more applications.

Better Test Reproducibility And Repeatability:

E **Constant Clamping Force in Lower Fabric Clamp (STA-0042)** to minimize operation error.

F **Minimum Clamping Failure & Slippage** chiefly because our **Universal Grip (STA-0041)** can accurately secure snap and does not damage the fabric.

Better Recognition:

G Unlike others, SafGuard has been adopted by all world's leading textile testing labs **(e.g. BV MTL_ACTS, Intertek, SGS & STR...)** & the most influential retailers / renowned brands **(e.g. GAP, H&M, & Ralph Lauren childrenswear...)**.

H SafGuard is Patented (No. 2005200005331.0)

Standard Package:

STA-0020
Imada Force Gauge FB-30K

Display the reading of pulling force with capacity up to 30 kg.



STA-0007
Upper Long Nose Vise Grip

Mainly used for trouser hooks and other smaller accessories.



STA-0041
Upper Universal Grip

Test for female part of snap & male part of stud.



STA-0002
Upper Stud Grip

Test for male part of baby snaps but generally Upper Universal Snap Grip can be also applied instead.



STA-0042
Lower Fabric Clamp, Level Arm Locking

Fix the sample with buttons firmly, flatly & constantly so that fabric slippage can be minimized.



STA-0006
Lower Grasp Button Accessory Kit

Convert the original lower fabric clamp for testing the grasp buttons or larger buttons to 30 ligne and it is mainly used together with STA-0005.



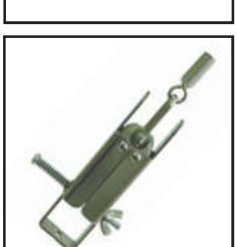
STA-0004
Calibration Weight 15LB

Routine checking the acuracy of the Imada Force Gauge.
(15LB = 6.82kg)



STA-0005
Upper Grasp Button Grip

Designed to gripping the grasp buttons or larger buttons, e.g. Jeans or Plastic Buttons up to 30 ligne.



Optional Accessories:

STA-0084
Multifunction Vise Clamp

Fix the garment between the jaw faces and will use with other upper grip for testing other attachments such as Patch, Side Tab, Zipper Puller, etc.



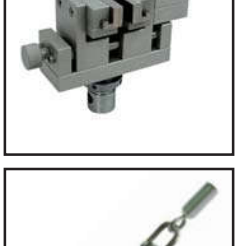
STA-0037
Small Part Cylinder - Plastic

To test whether the garment accessories are classified as "Small Part" or not in accordance with EN71 & ASTM F963



STA-0008
Three Pronged Grip

Test for four-hole buttons & Jeans buttons.



STA-0035
Goggles

To protect the eye of operator.



Notes:

- **a** Snapping & unsnapping (ASTM D4846) can be done by SafGuard by reverting Imada force gauge for compression.
- **b** For changing market needs, we have other special clamps, e.g. eyelets.
- **c** Software driven SafGuard for constant force & speed programmable for more advanced tests can be available upon request (e-mail: hcgartex&gar-tex.biz.com.hk).
- **d** Universal Grip (STA-0041) & Lower Fabric Clamp (STA-0042) can be used with other similar testers or even with UTM.
- **e** Other than standard package, there are several models for different requirements of major retailers & different application.

Professional Custom Engineering Services:

To keep up with changing market needs, professional custom designed clamps can be offered upon your request to use together with the apparatus. For quotation, please send us your end-product as samples, preliminary tests, and your specific requirements.

Packing Details:

Length x Width x Height :
98 x 39 x 46 (cm), Gross Weight: 48kg